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Title: Integrating Machine Learning for Sustaining Cybersecurity in Digital Banks

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Summary:

This paper examines the integration of machine learning (ML) algorithms into cybersecurity strategies for digital banks, addressing the escalating threats in an era of rapid technological advancement and widespread digital services. The authors provide an overview of digital banking, distinguishing its unique security challenges from traditional banking, such as heightened vulnerability to cyber threats due to online operations and vast data handling. Key cyber risks identified include phishing attacks, ransomware, distributed denial-of-service (DDoS) assaults, insider threats, and social engineering, which exploit human and technological weaknesses to compromise sensitive financial data and customer trust. The study emphasizes the need for robust, adaptive defenses to protect clients and maintain operational integrity, positioning ML as a pivotal tool for enhancing detection, prevention, and response capabilities in this dynamic environment.

The methodology involves a review of relevant ML algorithms tailored to cybersecurity in digital banking, including support vector machines (SVM) for anomaly detection, recurrent neural networks (RNN) for sequence-based threat analysis, hidden Markov models (HMM) for modeling probabilistic behaviors in fraud detection, and local outlier factor (LOF) for identifying unusual patterns in transaction data. The authors propose a comprehensive model for building a cybersecurity framework that incorporates ethical considerations, such as bias mitigation, data privacy, and transparency, to address potential vulnerabilities like algorithmic errors or unfair profiling. This framework aims to ensure that ML implementations are not only effective but also responsible, preventing unintended harms while fortifying systems against evolving threats.

Through a SWOT analysis, the paper outlines the strengths (e.g., real-time threat detection and scalability), weaknesses (e.g., dependency on high-quality data and computational demands), opportunities (e.g., proactive risk management and innovation in fraud prevention), and threats (e.g., adversarial attacks on ML models and regulatory compliance issues) of ML integration. The findings underscore ML's potential to bolster cybersecurity procedures, sustain digital bank operations, and foster customer confidence. However, the authors caution about drawbacks like implementation costs and the need for skilled personnel, advocating for balanced adoption strategies. Overall, the study contributes to understanding how ML can transform cybersecurity in digital banking, calling for ongoing research into ethical and practical enhancements to safeguard the ecosystem.